

Computational Methods for Geotechnical Sieve Analysis

Calculation Approaches and Validity Conditions
of the GSD Analysis Pipeline (ASTM D6913)

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Abstract

This document describes the computational methods implemented in the grain-size distribution (GSD) analysis pipeline for geotechnical sieve testing conducted in accordance with **ASTM D6913** [2]. All formulas, interpolation procedures, validity conditions, and screening-level estimation methods are presented in full. The pipeline consists of two scripts: `sieve_analysis_test.py`, which processes raw sieve-mass data and computes geotechnical parameters, and `plot_stratigraphy.py`, which visualises those parameters as a stratigraphic vertical profile. Where standard measurement fails due to high fines content ($> 10\%$ passing the #200 sieve), log-linear extrapolation and the Kozeny–Carman equation [4, 7] are used to provide indicative screening estimates that must be confirmed by hydrometer analysis [3].

1 Introduction

The primary objective of this document is to provide a transparent, traceable account of every mathematical operation performed by the GSD pipeline. It is *not* a site-specific results report; those are generated separately in `Site_N/report/`.

The pipeline follows a two-stage architecture decoupled by JSON:

1. **Stage 1 (`sieve_analysis_test.py`)**: Reads raw CSV sieve data, computes the cumulative percent-passing curve, interpolates characteristic grain diameters, and exports a GSD plot and a JSON report per sample.
2. **Stage 2 (`plot_stratigraphy.py`)**: Reads the JSON reports for all samples at a site and generates a two-panel stratigraphic profile of uniformity coefficient and hydraulic conductivity versus depth.

Governing standards referenced throughout this document:

- **ASTM D6913** – Particle-Size Distribution Using Sieve Analysis [2]
- **ASTM D7928** – Particle-Size Distribution Using Hydrometer Analysis [3]
- **ASTM D2487** – Unified Soil Classification System (USCS) [1]

2 Data Input and Pre-Processing

2.1 Input Format

Each sample is supplied as a two-column CSV file with columns `Sieve` and `Sample_Mass(g)`. Thirteen sieve designations are recognised; their physical opening sizes (the values marked on the laboratory sieve frames) are listed in Table 1.

Table 1: Sieve designations and physical opening sizes used in the pipeline.

Designation	Opening (mm)	Designation	Opening (mm)
#4	4.760	#100	0.149
#10	2.000	#140	0.105
#20	0.840	#200	0.074
#30	0.590	#400	0.037
#40	0.420	Pan	—
#50	0.297		
#60	0.250		
#80	0.177		

The Pan row acts as a catch-all; its placeholder size (0.001 mm) is used only for mass accounting and is *excluded* from interpolation (see Section 3).

2.2 Cumulative Percent Passing

Following ast [2], the data are sorted from coarsest to finest sieve. The cumulative mass retained at each sieve i is

$$M_{\text{cum},i} = \sum_{j=1}^i m_j \quad (1)$$

where m_j is the mass retained on sieve j (in grams) and the sum proceeds from the coarsest sieve downward. The percent passing is then

$$P_i = 100 \times \left(1 - \frac{M_{\text{cum},i}}{M_{\text{total}}} \right) \quad (2)$$

where $M_{\text{total}} = \sum_j m_j$ is the total dry specimen mass including the Pan. This yields a monotonically decreasing sequence $P_1 \geq P_2 \geq \dots \geq P_{13}$ (since the finest sieve always retains the remaining fines above the Pan).

3 Grain-Size Interpolation via PCHIP

3.1 Motivation for Log-Space Interpolation

Grain-size distribution curves are conventionally plotted on a semi-logarithmic axis (particle size on a log scale, percent passing on a linear scale). Interpolation performed in log-space therefore respects the shape of the curve and avoids the artefacts that linear-space interpolation introduces at the coarser end of the distribution.

3.2 PCHIP Method

The pipeline uses the *Piecewise Cubic Hermite Interpolating Polynomial* (PCHIP) algorithm [5], which guarantees monotonicity of the interpolant between successive data points. This is an important property for grain-size curves: spurious oscillations (overshoot) must be avoided because percent-passing values must remain in $[0, 100]$.

Formally, let $\{(P_i, d_i)\}$ be the set of measured (percent passing, grain diameter) pairs with the Pan excluded. PCHIP is applied to the log-transformed diameters:

$$D_x = 10^{\text{PCHIP}(x, \log_{10}(d))} \quad (3)$$

where x is the target percentile (e.g. $x = 10$ for D_{10}) and $\text{PCHIP}(\cdot)$ denotes the monotone cubic Hermite interpolant fitted to the pairs $\{(P_i, \log_{10} d_i)\}$.

3.3 Validity Conditions

Interpolation is strictly limited to the measured data range. If the target percentile x does not satisfy

$$\min_i(P_i) \leq x \leq \max_i(P_i), \quad (4)$$

the function raises a `ValueError` and the corresponding D_x value is reported as `null` in the JSON output. No extrapolation is performed within the measured range; when the curve cannot reach a percentile, a hydrometer analysis per ast [3] is required to extend the GSD curve into the fine-grained (silt and clay) domain.

3.4 Target Percentiles

The pipeline computes D_{10} , D_{25} , D_{30} , D_{50} , D_{60} , and D_{75} . D_{50} is used internally for the Kozeny–Carman estimate (Section 5.4) but is not reported in the public JSON output to avoid confusion with the directly computed characteristic diameters.

4 Measured Geotechnical Coefficients

The following coefficients are computed from the PCHIP-interpolated D_x values. Any coefficient whose prerequisite D_x is `null` (i.e. outside the measurable range) is itself reported as `null`.

4.1 Uniformity Coefficient C_u

$$C_u = \frac{D_{60}}{D_{10}} \quad (5)$$

C_u quantifies the spread of the grain-size distribution. A soil with $C_u = 1$ is perfectly uniform (single grain size); larger values indicate a broader, better-graded distribution. Per ast [1]:

- *Well-graded sand* (SW): $C_u \geq 6$ and $1 \leq C_c \leq 3$
- *Well-graded gravel* (GW): $C_u \geq 4$ and $1 \leq C_c \leq 3$

4.2 Coefficient of Curvature C_c

$$C_c = \frac{(D_{30})^2}{D_{60} \cdot D_{10}} \quad (6)$$

C_c describes the shape of the central portion of the GSD curve. The USCS well-graded criterion requires $1 \leq C_c \leq 3$ in conjunction with the C_u threshold stated above [1].

4.3 Trask Sorting Coefficient S_0

$$S_0 = \sqrt{\frac{D_{75}}{D_{25}}} \quad (7)$$

The Trask sorting coefficient [8] is a statistical measure of grain-size dispersion based on the geometric quartile deviation. Interpretation guidelines are summarised in Table 2.

Table 2: Trask sorting coefficient interpretation [8].

S_0 range	Classification
< 2.5	Well-sorted (uniform grains)
≈ 3	Normally sorted
> 4.5	Poorly sorted (wide grain-size range)

4.4 Hazen Hydraulic Conductivity K_{Hazen}

The empirical Hazen formula [6] relates hydraulic conductivity to the effective size D_{10} :

$$K_{\text{Hazen}} = (D_{10})^2 \quad [\text{cm/s}] \quad (8)$$

where the Hazen constant $C = 1.0$ is assumed (appropriate for medium, clean sands). The formula is applied only when *all* of the following conditions are satisfied:

1. D_{10} is within the measurable sieve range (not null),
2. $0.1 \text{ mm} \leq D_{10} \leq 3.0 \text{ mm}$ (effective-size validity window),
3. $C_u < 5$ (uniform-sand criterion for the Hazen approximation).

If any condition is unmet, K_{Hazen} is reported as null and the Kozeny–Carman estimate in Section 5.4 serves as a supplementary indicator.

5 Estimated Parameters for High-Fines Samples

5.1 Problem Statement

When more than approximately 10% of the sample passes the #200 sieve (0.074 mm), the effective grain diameter D_{10} falls below the measurable range of dry sieve analysis. Consequently, C_u , C_c , and K_{Hazen} cannot be computed from measurements alone.

ASTM D2487 acknowledges this limitation and permits extrapolation for the purpose of obtaining D_{10} [1]; however, any extrapolated value is indicative only and must be confirmed by hydrometer analysis [3].

5.2 Log-Linear D_{10} Extrapolation

When D_{10} is null, the pipeline extrapolates using the three sieve points with the lowest percent-passing values (i.e. the three finest physical sieves with measurable mass). In log-diameter versus percent-passing space, these points are approximated by a straight line:

$$\log_{10}(d_j) = m \cdot p_j + b, \quad j = 1, 2, 3 \quad (9)$$

$$D_{10,\text{est}} = 10^{m \cdot 10 + b} \quad (10)$$

where (m, b) are the least-squares slope and intercept obtained from `numpy.polyfit` applied to the three (percent-passing, log-diameter) pairs.

A validity guard is applied: the extrapolated value is accepted only if

$$0.001 \text{ mm} < D_{10,\text{est}} < 0.074 \text{ mm} \quad (11)$$

ensuring the estimate lies within the silt range (below the #200 sieve but above the Pan placeholder size). If the guard fails, $D_{10,\text{est}}$ is reported as `null`.

5.3 Estimated $C_{u,\text{est}}$ and $C_{c,\text{est}}$

Given a valid $D_{10,\text{est}}$, the measured D_{30} and D_{60} (which are typically available for samples with moderate-to-high fines) are substituted into the standard formulas:

$$C_{u,\text{est}} = \frac{D_{60}}{D_{10,\text{est}}} \quad (12)$$

$$C_{c,\text{est}} = \frac{(D_{30})^2}{D_{60} \cdot D_{10,\text{est}}} \quad (13)$$

These values are labelled *Estimated (indicative only) in both the JSON output and the GSD plot, and carry a warning directing the user to confirm with ASTM D7928.

5.4 Kozeny–Carman Hydraulic Conductivity K_{KC}

For samples where D_{50} is within the measurable range (i.e. less than 50% fines), the Kozeny–Carman equation [4, 7] provides a physics-based estimate of hydraulic conductivity:

$$K_{\text{KC}}(n) = \frac{g}{\nu} \cdot \frac{D_{50,\text{m}}^2}{180} \cdot \frac{n^3}{(1-n)^2} \times 100 \quad [\text{cm/s}] \quad (14)$$

where:

- $g = 9.81 \text{ m/s}^2$ – gravitational acceleration,
- $\nu = 1 \times 10^{-6} \text{ m}^2/\text{s}$ – kinematic viscosity of water at 20 °C,
- $D_{50,\text{m}} = D_{50} \times 10^{-3}$ – median grain diameter converted from mm to m,
- n – porosity (dimensionless).

Natural sandy soils typically exhibit porosities in the range $n \in [0.25, 0.50]$ [1]. The pipeline uses the narrower range $n \in [0.40, 0.45]$ representative of loose-to-medium dense sands, and reports two bounding estimates:

$$K_{\text{KC},\text{low}} = K_{\text{KC}}(n = 0.40) \quad (15)$$

$$K_{\text{KC},\text{high}} = K_{\text{KC}}(n = 0.45) \quad (16)$$

This bracketing approach communicates the sensitivity of the estimate to porosity uncertainty without requiring a direct measurement of void ratio.

6 Stratigraphic Profile Computation

The script `plot_stratigraphy.py` reads all Site 1 JSON reports and assembles a two-panel stratigraphic profile.

6.1 Depth Convention

Sample names follow the convention `Sample_1-H`, where H is the height in metres above the base of the 10 m excavation pit. The depth is extracted by parsing the integer suffix:

$$H = \text{int}(\text{sample_name.split('-')}[-1])$$

Thus $H = 0$ corresponds to the deepest (basal) layer and $H = 10$ to the surface layer.

6.2 Most-Permeable Layer Identification

To highlight the stratigraphic horizon of greatest hydraulic interest, the script identifies the layer with the maximum measured K_{Hazen} :

$$i^* = \arg \max_i K_{\text{Hazen},i} \quad (17)$$

This layer is annotated on both panels of the profile as *Most Permeable Layer. Note that samples with null K_{Hazen} (high-fines samples) are excluded from the $\arg \max$ operation via `numpy.nanargmax`.

6.3 Plot Layout

The output figure contains two subplots sharing the same y-axis (height above pit base):

- **Left panel:** C_u on a linear scale.
- **Right panel:** K_{Hazen} on a base-10 logarithmic scale (cm/s).

The y-axis is not inverted; $H = 0$ is at the bottom and $H = 10$ is at the top, matching the physical orientation of the pit.

7 Summary of Formulas and Validity Conditions

Table 3 consolidates all formulas, their equation numbers, and the conditions under which each quantity is computable.

Table 3: Summary of all formulas, references, and validity conditions.

Parameter	Formula	Ref.	Validity / Notes
P_i (percent passing)	(2)	[2]	Always computable
D_x (grain diameter)	(3)	[5]	$\min(P) \leq x \leq \max(P)$; Pan excluded
C_u (uniformity)	(5)	[1]	Requires D_{10} , D_{60}
C_c (curvature)	(6)	[1]	Requires D_{10} , D_{30} , D_{60}
S_0 (Trask sorting)	(7)	[8]	Requires D_{25} , D_{75}
K_{Hazen}	(8)	[6]	D_{10} available; 0.1 mm to 3.0 mm; $C_u < 5$
$D_{10,\text{est}}$	(10)	[1]	$D_{10} = \text{null}$; result in 0.001 mm to 0.074 mm
$C_{u,\text{est}}$	(12)	—	$D_{10,\text{est}}$ valid; confirm with ast [3]
$C_{c,\text{est}}$	(13)	—	$D_{10,\text{est}}$ valid; confirm with ast [3]
K_{KC} (low/high)	(14)	[4, 7]	D_{50} available; $n = 0.40\text{--}0.45$; indicative only
i^* (most permeable)	(17)	—	NaN-safe <code>nanargmax</code> over K_{Hazen}

References

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